

Elliptic curves

Discriminant: $p(x) = a_n x^n + a_{n-1} x^{n-1} + \dots + a_1 x + a_0$

Suppose $\alpha_1, \alpha_2, \dots, \alpha_n$ are the roots (some or all of them may be repeated).

$$D(p) = a_n^{2n-2} \times \prod_{1 \leq i < j \leq n} (\alpha_i - \alpha_j)^2$$

This is called the discriminant.

- A root of $p(x)$ is repeated if & only if $D(p) = 0$

- Recall $p(x) = ax^2 + bx + c$

$$\begin{aligned} D(p) &= a^2 (\alpha_1 - \alpha_2)^2 \\ &= a^2 \{ (\alpha_1 + \alpha_2)^2 - 4\alpha_1\alpha_2 \} \\ &= a^2 \left\{ \left(-\frac{b}{a}\right)^2 - 4\frac{c}{a} \right\} \\ &= \frac{b^2 - 4ac}{1} \end{aligned}$$

Exerc: if $p(x) = ax^3 + bx^2 + cx + d$, then

$$D(p) = \underline{b^2c^2} - 4\underline{b^3d} - 4\underline{ac^3} - 27\underline{a^2d^2} + 18\underline{abcd}$$

Then for $a=1, b=0$

$$\begin{aligned} D(x^3 + cx + d) &= -4c^3 - 27d^2 \\ &= -(4c^3 + 27d^2) \end{aligned}$$

$$\therefore \underline{D(x^3 + ax + b) = -(4a^3 + 27b^2)}$$

Cryptographic applications we use nonsingular

Elliptic curves over a finite field.

Characteristic, let F be a field

$$\mathbb{Z} \xrightarrow{\psi} F$$

$$1 \mapsto 1$$

$$n \mapsto n1 = \underbrace{(1 + \dots + 1)}_n$$

$$\text{if } n \geq 0$$

$$n1 = (-n)(-1) \quad \text{if } n < 0$$

• ~~$\mathbb{Z}$~~ $\psi(\mathbb{Z})$ integral domain

$$\& \text{ker } \psi \trianglelefteq \mathbb{Z} \quad \downarrow \text{ID}$$

$$\text{is s.t. } \frac{\mathbb{Z}}{\text{ker } \psi} \cong \psi(\mathbb{Z})$$

$\text{ker } \psi$ is a prime ideal

$$\text{so } \text{ker } \psi = \begin{cases} 0 \\ p\mathbb{Z} \end{cases}$$

p is a prime

so that $\frac{\mathbb{Z}}{p\mathbb{Z}}$ is a FD

$$\mathbb{C}, \mathbb{Q}, \mathbb{R} \leftarrow \begin{cases} \text{Char } F = 0 \\ \text{Char } F = p \end{cases}$$

$$\mathbb{Z}_p, \mathbb{F}_2, \mathbb{F}_{p^n}$$

$$\mathbb{Z}_p(x) = \left\{ \frac{f(x)}{g(x)} : f, g \in \mathbb{Z}_p[x] \right\}$$

In this case $\mathbb{Z}/p\mathbb{Z}$ is a field!

Defn: $n \in \mathbb{N}$ is the char

$$\text{of } F \text{ if } \underbrace{1+1+\dots+1}_n = 0$$

$$\text{but } \underbrace{1+1+\dots+1}_m \neq 0 \text{ for } 0 < m < n. \quad (\text{e.g. } \mathbb{Z}_p, n=p)$$

If no natural number is the char, then 0 is the char. of F . (i.e. $\mathbb{R}$)

We have shown that $\text{char}(F) = 0$ or p , a prime.

Nonsingular Elliptic curves for $\text{char}(\mathbb{F}) \neq 2, 3$

Run the standard for $\mathbb{F}$ an Elliptic curve that is nonsingular $\Rightarrow y^2 = x^3 + ax + b$ where $4a^3 + 27b^2 \neq 0$.

$$E = \{ (x, y) \in \mathbb{F}^2 \mid y^2 = x^3 + ax + b \} \cup \{\infty\}$$

where $4a^3 + 27b^2 \neq 0$.

For $\underline{p=2}$, $E = \{ (x, y) \in \mathbb{F}^2 \mid y^2 + cy = x^3 + ax + b \} \cup \{\infty\}$
 $x^3 + ax + b = 0$ has no repeated roots

$p=3$, $E = \{ (x, y) \in \mathbb{F}^2 \mid y^2 = x^3 + ax + b \} \cup \{\infty\}$
 no repeated root for $x^3 + ax + b$

or $E = \{ (x, y) \in \mathbb{F}^2 \mid y^2 = x^3 + ax^2 + b \} \cup \{\infty\}$

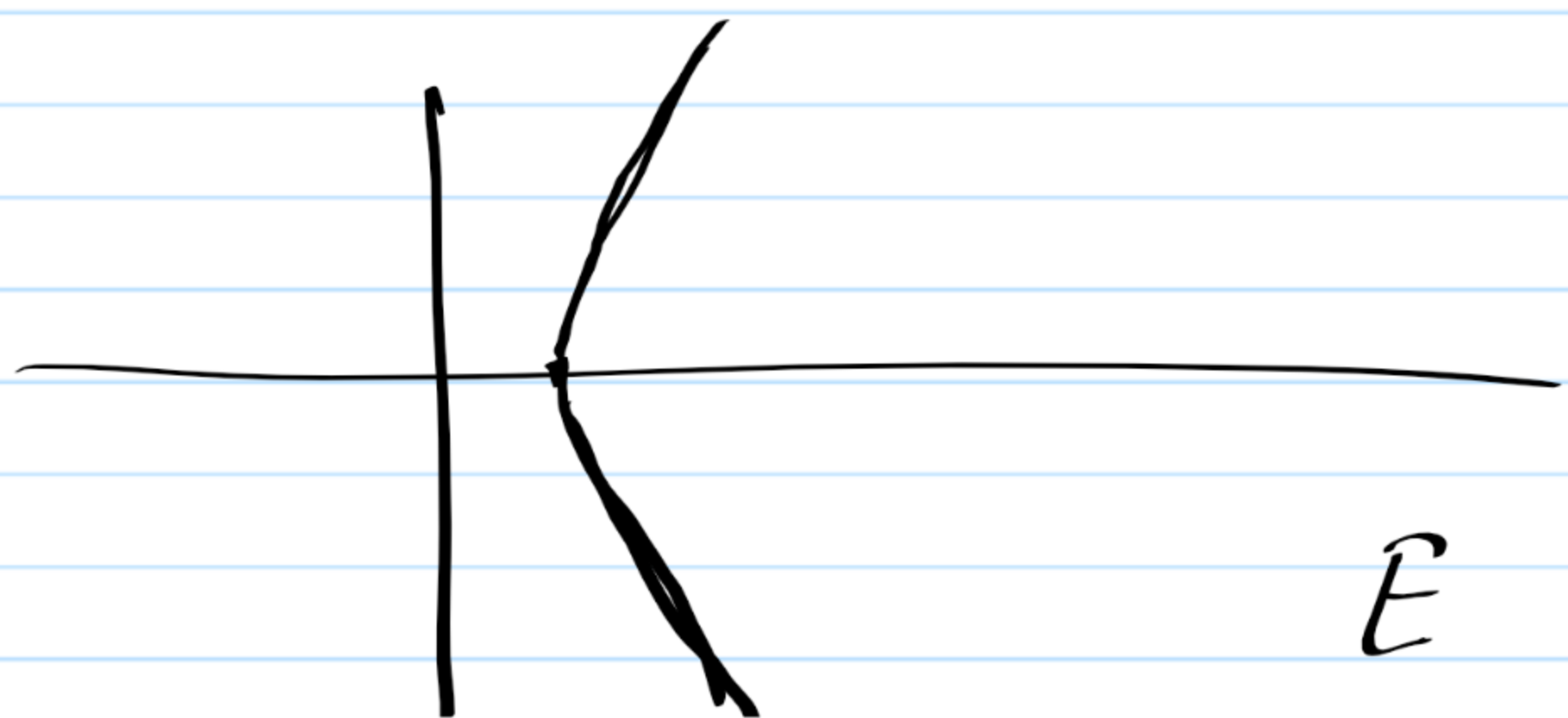
Remark $\Delta(E) = -16(4a^3 + 27b^2) \rightarrow$ is called the discriminant of E .
Graphs: Elliptic curves

Ex. 9. $E_1 = \{ (x, y) \in \mathbb{R}^2 \mid y^2 = x^3 + 2x - 3 \} \cup \{\infty\}$

$$\Delta(E_1) = -16 \{ 4 \cdot 2^3 + 27(-3)^2 \} < 0$$

$$y^2 = (x-1)(x^2 + x + 3)$$

no real roots



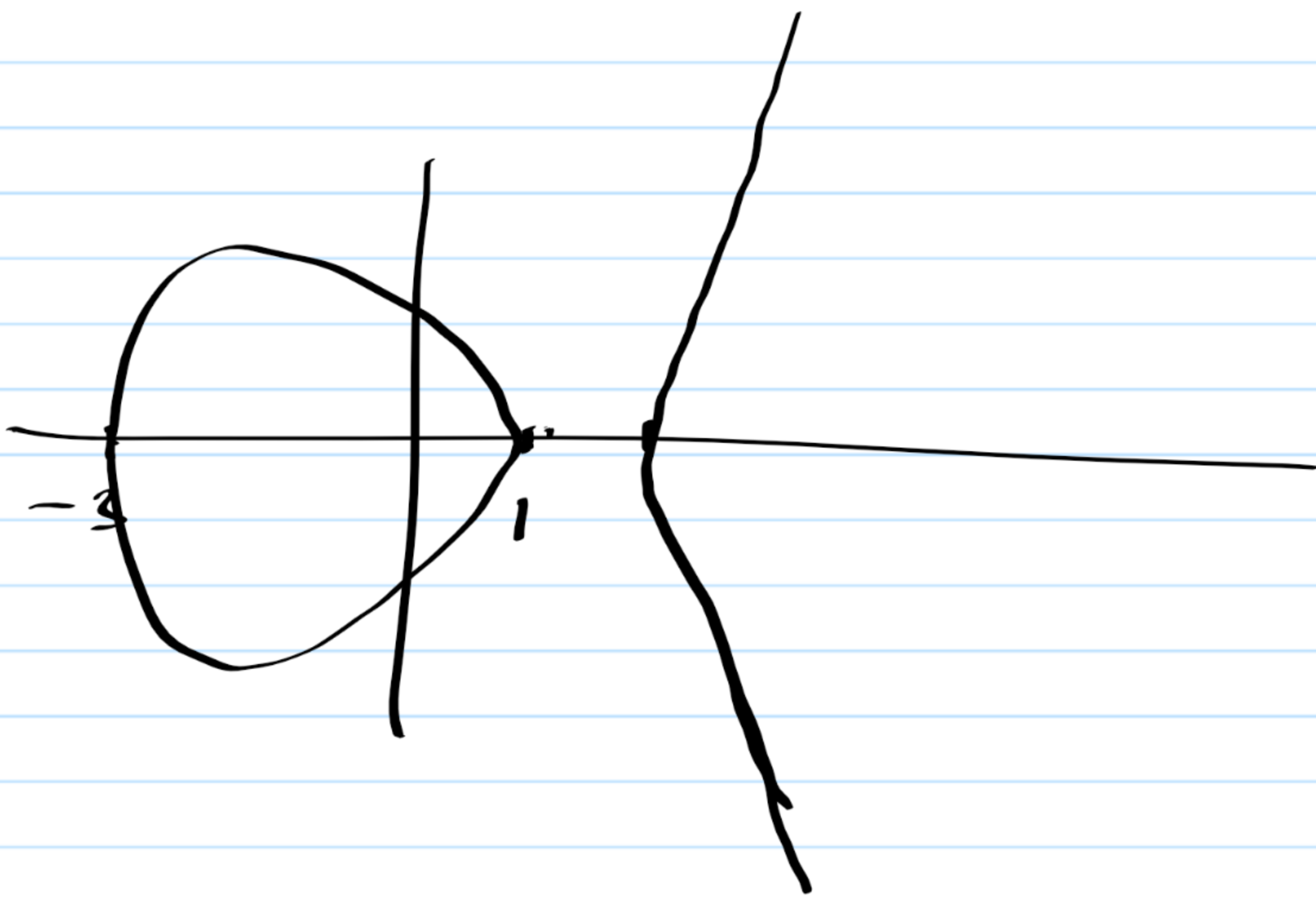
Remark: $x^3 + ax + b$ (cubic)
 has one real root & a pair of complex roots OR all real roots
 if & only if the discriminant is negative or positive simply.

$$E_2 = \{(x, y) \in \mathbb{R}^2 \mid y^2 = x^3 - 7x + 6\} \cup \{\infty\}$$

The discriminant is $-16 \times (4(-7)^3 + 27 \times 6^2) > 0$

So $x^3 - 7x + 6$ has three real roots.

In fact $y^2 = (x-1)(x-2)(x+3)$



Singular Elliptic curve

$$E_3 = \{(x, y) \in \mathbb{R}^2 \mid y^2 = x^3 + x^2\} \cup \{\infty\}$$

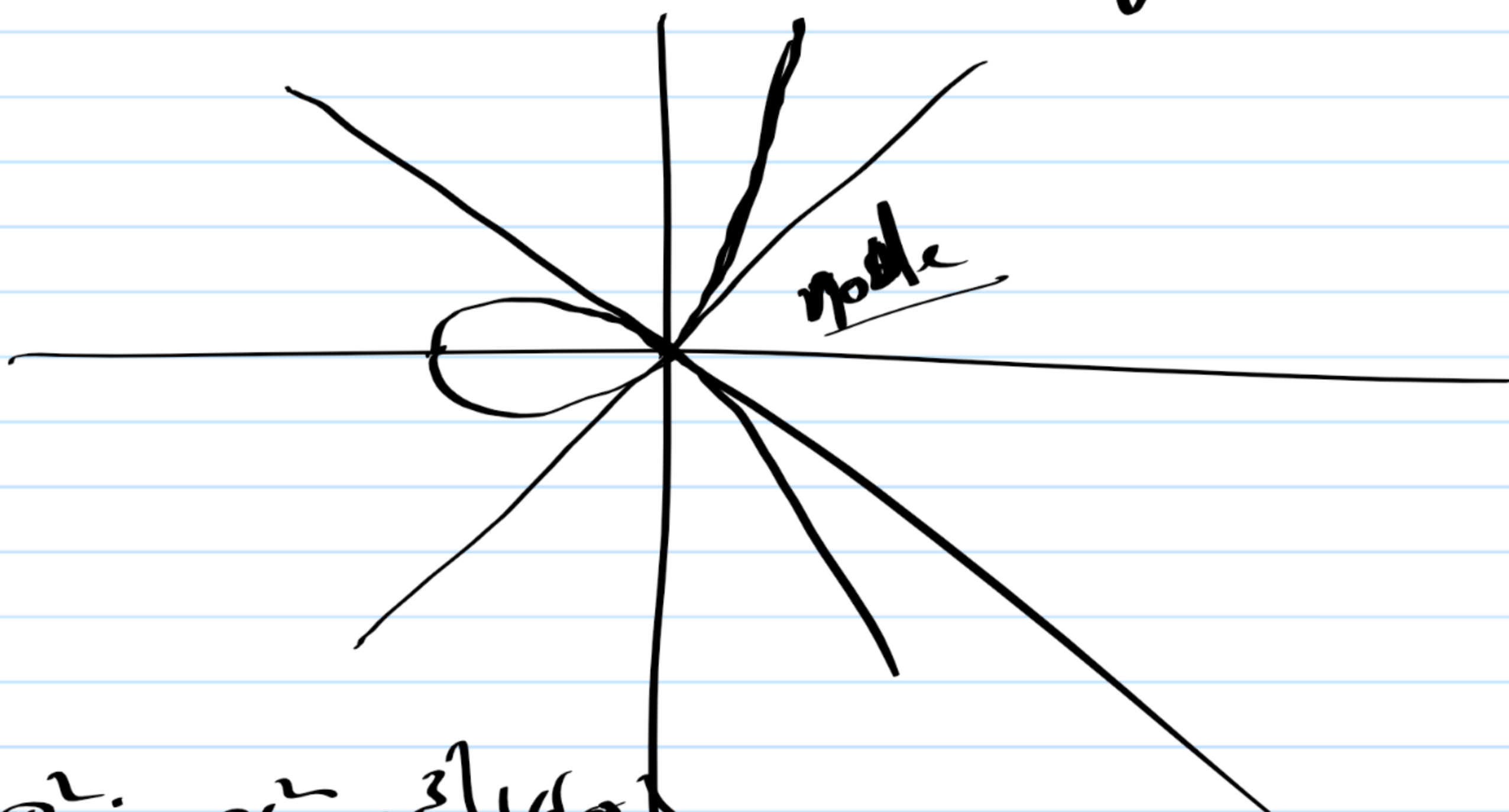
$(0, 0) \in E$ & tangents at $(0, 0)$ $y^2 = x^2$

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singular

i.e. $y = \pm x$

$$x^3 + x^2 = x^2(x+1) = 0$$

$$\Rightarrow x^2 = 0 \text{ or } x = -1$$



e.g., $E_4 = \{(x, y) \in \mathbb{R}^2 \mid y^2 = x^3\} \cup \{\infty\}$

$$x^3 = 0 \Rightarrow x = 0, 0, 0$$

Tangents: $y^2 = 0$ i.e. $y = 0$

